

The USD Isn't Worth the Cotton It's Printed On

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops an interdisciplinary examination of a deceptively simple proposition: “The USD isn’t worth the cotton it’s printed on.” The proposition is simultaneously a joke about the physical substrate of United States currency, a statement about fiat money, and an invitation to examine the relationship between nominal claims and real economic resources.

The analysis connects monetary economics, finance, econometrics, statistics, information theory, computer science, artificial intelligence, psychology, philosophy, political science, geopolitics, international relations, welfare economics, warfare economics, biology, chemistry, physics, and industrial organization. The central distinction is between the physical substrate of a monetary instrument, the nominal monetary claim represented by that instrument, and the institutional and economic system that sustains its purchasing power.

The paper also considers the extraordinary scale of contemporary financial claims by comparing the officially announced March 2026 post-money valuation of OpenAI with China’s foreign-exchange reserves and with the physical composition of United States banknotes. The resulting argument is not that money must be commodity-backed, but that monetary analysis becomes clearer when financial abstractions are continuously connected to information, institutions, production, resources, expectations, and real economic capacity.

The paper ends with “The End”

1 Introduction

A conversation about artificial intelligence, institutional epistemology, corporate valuation, foreign-exchange reserves, discount factors, and monetary economics eventually produced the following proposition:

$$\text{“The USD isn’t worth the cotton it’s printed on.”} \tag{1}$$

The proposition is funny because it is literally grounded in the physical construction of the United States banknote. The Bureau of Engraving and Printing states that United States currency paper is composed of approximately 75% cotton and 25% linen [1, 2].

The proposition is economically interesting because the monetary value of a banknote is not determined by the commodity value of its physical substrate. The distinction can be written as

$$V_{\text{note}} = V_{\text{monetary claim}} \neq V_{\text{substrate}}. \tag{2}$$

This inequality is a compact expression of a fundamental property of fiat money: the nominal value of the monetary instrument substantially exceeds the commodity value of the material from which the instrument is manufactured.

The purpose of this paper is therefore not to argue that the dollar should be cotton-backed. Rather, it is to examine what makes the monetary claim valuable when its physical substrate is worth comparatively little.

2 The Physical Currency

United States Federal Reserve notes are manufactured from a specialized substrate containing approximately 75% cotton and 25% linen [1, 2]. The Bureau of Engraving and Printing describes the substrate as a security technology rather than merely as ordinary paper.

Thus a banknote can be decomposed schematically as

$$N = S + I + C + L, \quad (3)$$

where S denotes the physical substrate, I denotes printed information, C denotes security and authentication characteristics, and L denotes the legal and monetary status associated with the instrument.

The decomposition is conceptual rather than additive in a literal accounting sense. Its purpose is to emphasize that the monetary object is a composite technological and institutional artifact.

The material component is real, but its monetary significance is not primarily determined by the market value of cotton or linen.

$$V_{\text{money}} \gg V_{\text{material}} \quad (4)$$

for ordinary circulating denominations.

3 Commodity Value and Fiat Value

Under a commodity-money system, the physical object can itself possess substantial value independently of its monetary designation. Under a fiat system, by contrast, the monetary value of the instrument depends on a broader economic and institutional structure.

Let

$$M = \text{nominal monetary stock}, \quad (5)$$

and let

$$P = \text{general price level}. \quad (6)$$

A simple representation of real monetary balances is

$$m = \frac{M}{P}. \quad (7)$$

The purchasing power of money is therefore a function of the relationship between nominal monetary quantities and the prices of goods and services, not of the quantity of cotton incorporated into the banknote.

The statement “the dollar is not worth the cotton it is printed on” is therefore compatible with fiat monetary theory.

4 Money as Information

Money can also be interpreted as an information system.

A monetary unit communicates a standardized claim within an economy. It facilitates exchange by reducing the informational requirements associated with barter.

Let agents i and j possess goods x_i and x_j . Under barter, exchange requires a sufficiently strong coincidence of wants. Under monetary exchange, the monetary instrument acts as an intermediate state variable:

$$x_i \rightarrow M \rightarrow x_j. \quad (8)$$

Money therefore performs at least three classical functions:

1. medium of exchange;
2. unit of account;
3. store of value.

These functions are institutionalized through law, banking systems, payment infrastructure, taxation, accounting conventions, and expectations.

Consequently,

$$\text{monetary value} = f(\text{institutions, expectations, liquidity, production, law, credibility}). \quad (9)$$

The physical substrate is necessary for a particular banknote but is not sufficient to explain its monetary value.

5 The Information-Theoretic Perspective

A monetary system compresses a large quantity of economic information into a standardized numerical representation.

Prices can be interpreted as signals.

$$p_i = f_i(\text{scarcity, preferences, technology, expectations, institutions}). \quad (10)$$

The monetary system permits heterogeneous goods and services to be represented using a common numerical language.

This creates a fundamental information transformation:

$$\text{heterogeneous real economy} \longrightarrow \text{price vector} \longrightarrow \text{monetary accounts}. \quad (11)$$

The transformation is enormously useful, but it is lossy. A single price does not contain every physical, social, ecological, strategic, or psychological property of the underlying object.

Thus:

$$\text{Price information} \neq \text{complete economic information}. \quad (12)$$

This observation connects monetary economics to the broader epistemological problem of data compression.

6 Institutional Epistemology

Institutions can stabilize expectations, coordinate transactions, and enforce contracts. They can also determine which information is recognized as legitimate.

An institutional claim should therefore be distinguished from a substantive argument:

$$\text{institutional provenance} \neq \text{logical validity}. \quad (13)$$

Likewise,

$$\text{institutional legitimacy} \neq \text{economic truth}. \quad (14)$$

This distinction is particularly relevant to central banks, financial institutions, statistical agencies, universities, rating agencies, technology companies, and governments.

Institutional status can increase credibility, reduce transaction costs, and coordinate expectations. It cannot logically guarantee that every proposition produced by the institution is true.

7 The AI Analogy

The same distinction applies to artificial intelligence.

An AI system can be represented schematically as

$$\text{World} \rightarrow \text{Data} \rightarrow \text{Representation} \rightarrow \text{Learning} \rightarrow \text{Inference} \rightarrow \text{Output}. \quad (15)$$

The paper *Introspection and Retrospection on the Age of Data Science, Neural Networks and AI* describes a modern data-science pipeline involving collection, cleaning, exploratory analysis, model building, deployment, and continuous feedback [5].

The corresponding computational transformation can be written as

$$\hat{y} = f_{\theta}(x), \quad (16)$$

where x represents input information, θ represents learned parameters, and $\hat{y}$ represents an output. For a neural network,

$$y = f(Wx + b). \quad (17)$$

For a Transformer attention mechanism,

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V. \quad (18)$$

These transformations do not reproduce the world directly. They construct representations of information.

Consequently, the appropriate epistemological question is not simply whether an AI system is “intelligent”, but

$$\boxed{\text{What information is retained, discarded, transformed, or reweighted?}} \quad (19)$$

8 Objective Functions

Machine-learning systems are optimized through objective functions.

A generic training problem may be represented as

$$\theta^* = \arg \min_{\theta} \mathcal{L}(\theta). \quad (20)$$

For supervised learning,

$$\mathcal{L} = \sum_{i=1}^n \ell(y_i, \hat{y}_i). \quad (21)$$

For reinforcement learning, a policy can be represented as

$$\pi(a|s) = \Pr(a_t = a | s_t = s), \quad (22)$$

with objective

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]. \quad (23)$$

The important distinction is between an optimization objective and an institutional objective.

$$\boxed{J_{\text{model}} \neq J_{\text{corporate}} \neq J_{\text{investor}}.} \quad (24)$$

The model may be optimized for a collection of behavioral objectives. The corporation may seek economic sustainability and growth. Investors may seek risk-adjusted returns.

There is no requirement that these objectives have identical optima.

9 OpenAI as an Economic Object

OpenAI provides a particularly interesting contemporary example.

On March 31, 2026, OpenAI announced a financing round involving 122 billion in committed capital at a post-money valuation of 852 billion [3].

This is a private-company valuation rather than a conventional public-market capitalization.

Thus

$$V_{\text{OpenAI}} \neq P_{\text{share}} \times N_{\text{public shares}}, \quad (25)$$

because there is no continuously traded public equity price of the ordinary kind.

The valuation is nevertheless economically meaningful because investors assign a present value to expected future economic benefits.

A stylized discounted-cash-flow representation is

$$V_0 = \mathbb{E}_0 \left[\sum_{t=1}^{\infty} \frac{CF_t}{(1+r_t)^t} \right]. \quad (26)$$

The valuation therefore represents expectations about future economic capacity rather than the physical value of the company’s assets alone.

10 The “Expensive Autocomplete” Hypothesis

An intentionally provocative hypothesis is:

$$H_0 : \text{A large language model is fundamentally a sophisticated conditional text-completion system.} \quad (27)$$

The corresponding economic question is different:

$$H_1 : \text{The system generates sufficient economic surplus to justify its capital requirements and risk.} \quad (28)$$

The two propositions are not mutually exclusive.

A system could have a relatively simple computational description and still generate enormous economic value.

Hence

$$\text{economic value} \neq \text{ontological sophistication.} \quad (29)$$

This distinction is fundamental to technology economics.

11 Risk-Adjusted Valuation

Investors are not concerned only with expected returns. They also consider risk.

The Sharpe ratio is conventionally defined as

$$S = \frac{\mathbb{E}[R_p - R_f]}{\sigma(R_p - R_f)}. \quad (30)$$

For a private company, a conventional Sharpe ratio is problematic because there may be no continuous public-market return series.

If private valuations V_t are observed periodically, a valuation-return proxy can be constructed as

$$R_t^* = \frac{V_t - V_{t-1}}{V_{t-1}}. \quad (31)$$

A corresponding proxy Sharpe statistic is

$$S^* = \frac{\overline{R^*} - \overline{R_f}}{s(R^*)}. \quad (32)$$

However, S^* is not a conventional traded-equity Sharpe ratio. It is subject to valuation-frequency, liquidity, stale-price, selection, and measurement problems.

This is a standard econometric warning:

$$\text{observed valuation} \neq \text{frictionless market price.} \quad (33)$$

12 China’s Foreign-Exchange Reserves

China provides another useful scale comparison.

The State Administration of Foreign Exchange reported foreign-exchange reserves of approximately

$$R_{\text{China}} = \$3.4383 \text{ trillion} \quad (34)$$

at the end of August 2026, an increase of approximately \$19.5 billion from July [4].

The comparison

$$\frac{3.4383}{0.852} \approx 4.04 \quad (35)$$

shows that China’s foreign-exchange reserves were approximately four times OpenAI’s officially announced March 2026 post-money valuation.

This comparison is purely dimensional and should not be interpreted as saying that foreign-exchange reserves and corporate equity valuations are equivalent economic objects.

Indeed,

$$\boxed{\text{reserve asset} \neq \text{corporate equity claim.}} \quad (36)$$

The comparison is useful precisely because the objects are different.

13 Currency, Capital, and Scale

Large dollar-denominated quantities can refer to very different economic objects:

Quantity	Economic object	Liquidity characteristic
Banknote	Monetary instrument	Very high
FX reserve	Reserve asset portfolio	High
Corporate valuation	Equity valuation claim	Variable
Future cash flow	Expected income stream	Uncertain
Physical cotton	Commodity input	Market-dependent

Consequently, adding dollar-denominated quantities is not generally an economically meaningful aggregation.

For example,

$$\$852 \text{ billion} + \$3.4383 \text{ trillion} \quad (37)$$

is mathematically valid as a dimensional operation, but it does not create a new monetary aggregate with an obvious economic interpretation.

The correct question is:

$$\boxed{\text{What claims do these quantities represent?}} \quad (38)$$

14 Discount Factors and Valuation

The reciprocal of a valuation,

$$\frac{1}{V}, \quad (39)$$

is not itself a conventional discount factor because it has units of inverse currency.

A standard discrete discount factor is

$$D_t = \frac{1}{(1+r)^t}. \quad (40)$$

Under continuous compounding,

$$D_t = e^{-rt}. \quad (41)$$

A dimensionless relative valuation factor can instead be constructed as

$$D_V(t) = \frac{V_0}{V_t}. \quad (42)$$

This distinction illustrates a general principle of dimensional analysis:

$$\boxed{\text{mathematical reciprocity} \neq \text{economic discounting.}} \quad (43)$$

15 Cotton as an Economic System

The paper *Cotton Technology* explicitly treats cotton as a multidisciplinary technological system spanning agriculture, fiber processing, manufacturing, logistics, commodity finance, international trade, military procurement, data-driven optimization, and artificial intelligence [6].

The integrated value chain is represented as

$$\text{Cultivation} \rightarrow \text{Harvesting} \rightarrow \text{Ginning} \rightarrow \text{Spinning} \rightarrow \text{Textiles} \rightarrow \text{Retail}. \quad (44)$$

Cotton therefore provides a useful reminder that apparently simple commodities can embody enormous technological and institutional systems.

A stylized production function is

$$Y = AN^\alpha W^\beta M^\gamma, \quad (45)$$

where N denotes nutrient input, W water input, M machinery intensity, and A a technology factor [6].

Total factor productivity can be written as

$$TFP = \frac{Y}{K^\alpha L^\beta}. \quad (46)$$

Thus even cotton can be analyzed as a technologically mediated capital, labor, resource, and information system.

16 Chemistry and Materials Science

Cotton is primarily a cellulose-based natural fiber. At the molecular level, cellulose consists of repeating glucose-derived units connected through $\beta(1 \rightarrow 4)$ glycosidic linkages.

A simplified empirical representation is

$$(C_6H_{10}O_5)_n. \quad (47)$$

The chemical composition alone, however, does not determine the economic value of cotton.

Mechanical properties, fiber length, diameter, orientation, crystallinity, moisture, processing quality, and market demand all matter.

A simplified mechanical relationship is

$$\sigma = E\epsilon, \quad (48)$$

where σ is stress, E Young's modulus, and ϵ strain [6].

This demonstrates another general principle:

$$\text{material composition} \neq \text{economic value}. \quad (49)$$

17 Physics and Dimensional Analysis

Physics provides an important methodological constraint.

A physical quantity must have a dimensionally coherent representation.

For example,

$$[V] = [\text{USD}], \quad (50)$$

while

$$\left[\frac{1}{V}\right] = [\text{USD}^{-1}]. \quad (51)$$

A conventional discount factor satisfies

$$[D_t] = 1. \quad (52)$$

Dimensional analysis therefore prevents the careless identification of $1/V$ with a discount factor. The same principle applies to economics.

$$\boxed{\text{Dimensional consistency is necessary but not sufficient for economic interpretation.}} \quad (53)$$

18 Statistics and Econometrics

Suppose an analyst observes a valuation sequence V_t .

The first transformation is

$$R_t = \frac{V_t}{V_{t-1}} - 1. \quad (54)$$

The sample mean is

$$\bar{R} = \frac{1}{T} \sum_{t=1}^T R_t. \quad (55)$$

The sample variance is

$$s^2 = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2. \quad (56)$$

However, sparse private valuations introduce several econometric problems:

1. irregular observation intervals;
2. stale valuation marks;
3. transaction-selection bias;
4. heterogeneous investor rights;
5. non-synchronous observations;
6. endogenous valuation events;
7. absence of continuous price discovery.

Consequently, statistical inference must distinguish between an observed valuation and an equilibrium market price.

19 Psychology and Behavioral Economics

Money has a psychological dimension because monetary value depends partly upon expectations.

Let

$$E_t[P_{t+1}] \quad (57)$$

denote an agent's expectation of the future price level.

Expectations influence current behavior:

$$\text{expectations} \rightarrow \text{spending} \rightarrow \text{prices} \rightarrow \text{future expectations}. \quad (58)$$

This creates a feedback system.

Behavioral economics further recognizes that people do not always process probabilities, uncertainty, and institutional signals according to the assumptions of a frictionless rational-agent model.

The monetary system therefore contains both material and psychological components.

20 Psychiatry and the Limits of Metaphor

Psychiatric language should be used cautiously in economic and technological analysis.

Institutions, markets, corporations, and algorithms are sometimes described metaphorically as “irrational,” “delusional,” “obsessed,” or “insane.” Such terminology should not be treated as clinical diagnosis.

A rigorous distinction is

$$\text{behavioral anomaly} \neq \text{psychiatric disorder.} \quad (59)$$

This matters because interdisciplinary analysis should not convert technical or institutional behavior into unsupported claims about individual mental states.

21 Philosophy

The central philosophical distinction can be stated as

$$\text{representation} \neq \text{reality.} \quad (60)$$

A banknote represents monetary value.

A valuation represents expected economic value.

A statistical model represents observed relationships.

A neural network represents learned patterns.

An institution represents a socially recognized structure of authority, coordination, or enforcement.

None of these representations is identical to the underlying reality.

This is the epistemological core of the paper.

22 Religion and Symbolic Value

Religious systems provide another illustration of symbolic value.

A physical object can acquire significance that exceeds its material composition through collective recognition, ritual, tradition, and shared belief.

The analytical point is not theological equivalence between money and religion. Rather, it is that humans routinely assign socially coordinated meaning to physical objects.

Thus:

$$\text{physical object} + \text{collective recognition} \rightarrow \text{symbolic value.} \quad (61)$$

Money is one of the most technologically consequential examples of this general phenomenon.

23 Political Science and Sovereignty

Currency also has a political dimension.

The ability to define legal tender, levy taxes, regulate banking, and operate payment infrastructure is closely associated with state capacity.

A simplified sovereign monetary system may be represented as

$$\mathcal{S} = \{\text{law, taxation, central banking, banking, payments, enforcement}\}. \quad (62)$$

The credibility of the monetary system therefore depends partly on the credibility of the political system supporting it.

This does not imply that political institutions are always correct. It implies that monetary systems are embedded within political structures.

24 Geopolitics and International Relations

A currency can acquire international importance when foreign governments, banks, corporations, and investors voluntarily or institutionally use it for trade, reserves, finance, and settlement.

A stylized international currency network can be represented by a weighted graph

$$G = (V, E), \quad (63)$$

where nodes represent economies and edges represent financial or trade relationships.

The importance of a currency may then depend upon network effects:

$$U_i = f(L_i, N_i, S_i, C_i), \quad (64)$$

where L_i denotes liquidity, N_i network usage, S_i settlement infrastructure, and C_i credibility.

This creates positive feedback:

$$\text{international use} \rightarrow \text{liquidity} \rightarrow \text{lower transaction costs} \rightarrow \text{greater international use}. \quad (65)$$

25 Diplomacy and International Settlement

International diplomacy operates partly through economic claims.

Trade agreements, sanctions, reserves, correspondent banking, payment systems, and sovereign debt all interact with monetary power.

The currency system is consequently part of international bargaining infrastructure.

Nevertheless,

$$\text{currency power} \neq \text{complete geopolitical power}. \quad (66)$$

Military capability, productive capacity, demographics, technology, alliances, geography, and diplomatic credibility also matter.

26 Welfare Economics

The social value of money is not merely its private purchasing power.

A stable monetary system can reduce transaction costs and facilitate specialization.

Let social welfare be

$$W = W(C, L, S, I), \quad (67)$$

where C represents consumption, L labor outcomes, S stability, and I investment.

Monetary instability can impose welfare costs through inflation uncertainty, redistribution, unemployment, financial distress, and reduced investment.

Therefore the welfare question is not simply

$$\text{What is a dollar made of?} \quad (68)$$

but

$$\boxed{\text{What economic coordination does a dollar enable?}} \quad (69)$$

27 Warfare Economics

War demonstrates that monetary systems ultimately depend upon productive capacity.

Military capability requires resources:

$$\text{military capability} = f(\text{labor, capital, energy, materials, technology, logistics}). \quad (70)$$

The *Cotton Technology* paper explicitly identifies military procurement and supply resilience as components of the cotton system [6].

A stylized wartime supply-resilience measure is

$$R = \frac{S}{D}, \quad (71)$$

where S is available supply and D military demand.
If

$$R > 1, \quad (72)$$

supply is sufficient under the model's assumptions.
This demonstrates the ultimate limitation of financial abstraction:

money can command resources, but money is not the resources themselves.

(73)

28 Computer Science and Algorithms

Economic systems increasingly resemble information-processing systems.

A simplified financial-processing algorithm is:

INPUT:

```
prices
quantities
inventories
expectations
institutional constraints
```

TRANSFORM:

```
clean observations
construct variables
estimate parameters
calculate risks
generate forecasts
```

OUTPUT:

```
prices
allocations
investment decisions
policy decisions
```

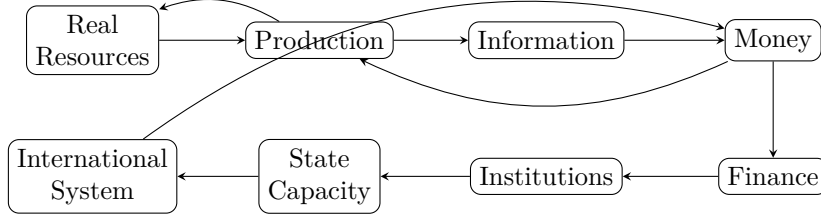
A corresponding pseudocode representation is

```
Algorithm EconomicInformationSystem
  Collect economic observations
  Validate and clean observations
  Construct state vector X_t
  Estimate model parameters theta
  Compute expected outcomes
  Compute risk measures
  Update decisions
  Observe realized outcomes
  Feed observations into next iteration
End Algorithm
```

The system therefore resembles a feedback controller:

$$X_{t+1} = F(X_t, U_t, \epsilon_t), \quad (74)$$

where X_t is the state, U_t the control or decision vector, and ϵ_t an innovation.



29 Data Science and AI/ML

Modern data science introduces another layer:

$$D \rightarrow \phi(D) \rightarrow M_\theta \rightarrow \hat{Y} \rightarrow A, \quad (75)$$

where D denotes raw data, ϕ feature transformation, M_θ a learned model, $\hat{Y}$ predictions, and A actions.

The transformation can be lossy.

$$I(D; \phi(D)) \leq H(D), \quad (76)$$

where $H(D)$ represents information content and I mutual information.

The exact inequality is a conceptual information-theoretic representation: a representation cannot automatically be assumed to preserve every feature of the source.

The same principle applies to monetary prices.

$$\text{economic reality} \rightarrow \text{prices} \quad (77)$$

is a representation, not a complete copy of economic reality.

30 An Integrated Systems View

The entire argument can be represented by the following conceptual architecture.

The arrows should not be interpreted as a complete causal model. They represent feedback relationships between real resources, production, information, finance, institutions, states, and international systems.

31 The Cotton–Dollar Paradox

The central paradox can now be expressed precisely.

A dollar banknote has a physical substrate with substantial cotton content:

$$C_{\text{banknote}} \approx 0.75. \quad (78)$$

But its monetary value is not approximately 0.75 units of cotton.

Instead,

$$V_{\text{USD}} = f(\text{legal status}, \text{tax system}, \text{monetary policy}, \text{banking system}, \text{liquidity}, \text{expectations}, \text{productive capacity}, \text{international demand}). \quad (79)$$

Therefore,

$$V_{\text{USD}} \not\approx V_{\text{cotton substrate}}. \quad (80)$$

The physical substrate is necessary for the manufactured banknote but economically secondary to the monetary system represented by the note.

32 The Broader Lesson

The same structure appears repeatedly throughout modern economics.

Table 2: Representation versus underlying reality

Representation	Underlying object	Potential information loss
Banknote	Monetary system	Physical and institutional abstraction
Price	Good or service	Non-price attributes
GDP	Economy	Distribution and non-market activity
Valuation	Enterprise	Physical and organizational capacity
Model	Data-generating process	Specification error
AI output	Information environment	Context and uncertainty
Institution	Social structure	Heterogeneous individual behavior

The common problem is therefore not merely monetary.

It is a general problem of abstraction:

$$\boxed{\text{Representation} \rightarrow \text{compression} \rightarrow \text{usefulness} + \text{information loss.}} \quad (81)$$

The appropriate analytical response is neither to reject abstraction nor to treat the abstraction as reality.

33 Conclusion

The sentence “The USD isn’t worth the cotton it’s printed on” is literally correct in the narrow material sense and economically revealing in the broader sense.

The United States banknote is a physical object whose substrate contains approximately 75% cotton and 25% linen. Yet its monetary value derives primarily from the economic, legal, political, financial, technological, and international system in which it circulates.

Similarly, a corporate valuation is not the same thing as physical productive capacity. A statistical variable is not the entire economy. A model is not the data-generating process. An AI output is not the world.

The general principle is therefore:

$$\boxed{\text{Nominal representation} \neq \text{underlying reality.}} \quad (82)$$

But the distinction should not be interpreted as saying that representations are useless. On the contrary, representations are among civilization’s most powerful technologies.

The dollar coordinates economic activity.

Prices coordinate decentralized decisions.

Financial markets aggregate dispersed information.

Institutions coordinate collective action.

Statistical models compress observations.

Algorithms automate transformations.

Artificial intelligence transforms information into predictions and actions.

The intellectual task is therefore to maintain awareness of the mapping:

$$\boxed{\text{Reality} \rightarrow \text{Representation} \rightarrow \text{Decision} \rightarrow \text{Outcome} \rightarrow \text{Feedback.}} \quad (83)$$

The joke about cotton therefore contains a serious methodological lesson.

Before asking how much a representation is worth, one should ask what the representation represents.

Before treating a valuation as wealth, one should identify the underlying claims.

Before treating a model as reality, one should identify the information it discarded.

And before constructing an elaborate theory about X , one should first make sure one knows what X is.

$$\boxed{\text{The cotton was there all along.}} \quad (84)$$

References

- [1] U.S. Bureau of Engraving and Printing. *Currency FAQs*. United States Department of the Treasury.
- [2] U.S. Bureau of Engraving and Printing. *The Buck Starts Here: How Money is Made*. United States Department of the Treasury.
- [3] OpenAI. “OpenAI raises \$122 billion to accelerate the next phase of AI.” March 31, 2026.
- [4] State Administration of Foreign Exchange. “SAFE Releases Data on China’s Foreign Exchange Reserve at the End of August 2026.” September 7, 2026.
- [5] Ghosh, S. *Introspection and Retrospection on the Age of Data Science, Neural Networks and AI: A Synthesis of Historical Development and Future Trajectories*. 2026.
- [6] Ghosh, S. *Cotton Technology: Intricacies and Applications*. 2026.
- [7] Keynes, J. M. *The General Theory of Employment, Interest and Money*. Macmillan, 1936.
- [8] Hayek, F. A. “The Use of Knowledge in Society.” *American Economic Review*, 35(4), 1945, 519–530.
- [9] Friedman, M. *The Quantity Theory of Money: A Restatement*. In: Studies in the Quantity Theory of Money. University of Chicago Press, 1956.
- [10] Lucas, R. E. “Econometric Policy Evaluation: A Critique.” In: Brunner, K. and Meltzer, A. H. (eds.), *The Phillips Curve and Labor Markets*. North-Holland, 1976.
- [11] Arrow, K. J. and Debreu, G. “Existence of an Equilibrium for a Competitive Economy.” *Econometrica*, 22(3), 1954, 265–290.
- [12] Shannon, C. E. “A Mathematical Theory of Communication.” *Bell System Technical Journal*, 27, 1948, 379–423, 623–656.
- [13] Markowitz, H. “Portfolio Selection.” *Journal of Finance*, 7(1), 1952, 77–91.
- [14] Sharpe, W. F. “Mutual Fund Performance.” *Journal of Business*, 39(1), 1966, 119–138.
- [15] Black, F. and Scholes, M. “The Pricing of Options and Corporate Liabilities.” *Journal of Political Economy*, 81(3), 1973, 637–654.
- [16] Engel, J., Hays, S. and others. Standard treatments of commodity production, agricultural economics, and industrial organization.
- [17] Turing, A. M. “Computing Machinery and Intelligence.” *Mind*, 59(236), 1950, 433–460.
- [18] Rosenblatt, F. “The Perceptron: A Probabilistic Model for Information Storage and Organization in the Brain.” *Psychological Review*, 65(6), 1958, 386–408.
- [19] Rumelhart, D. E., Hinton, G. E. and Williams, R. J. “Learning Representations by Back-Propagating Errors.” *Nature*, 323, 1986, 533–536.
- [20] Vaswani, A., Shazeer, N., Parmar, N., et al. “Attention Is All You Need.” *Advances in Neural Information Processing Systems*, 30, 2017.
- [21] Goodfellow, I., Bengio, Y. and Courville, A. *Deep Learning*. MIT Press, 2016.
- [22] Russell, S. and Norvig, P. *Artificial Intelligence: A Modern Approach*. 4th ed., Pearson, 2021.
- [23] Bollerslev, T. “Generalized Autoregressive Conditional Heteroskedasticity.” *Journal of Econometrics*, 31(3), 1986, 307–327.
- [24] Mas-Colell, A., Whinston, M. D. and Green, J. R. *Microeconomic Theory*. Oxford University Press, 1995.
- [25] North, D. C. *Institutions, Institutional Change and Economic Performance*. Cambridge University Press, 1990.
- [26] Ostrom, E. *Governing the Commons*. Cambridge University Press, 1990.

Glossary

Artificial Intelligence

Computational systems designed to perform tasks associated with perception, reasoning, learning, prediction, generation, or decision-making.

Banknote

A physical monetary instrument representing a monetary claim and incorporating security, authentication, and legal-tender characteristics.

Commodity Money

Money whose physical commodity has substantial value independently of its monetary function.

Discount Factor

A dimensionless factor used to transform future values into present values.

Fiat Money

Money whose value is not determined by a fixed commodity-conversion rule but depends upon monetary, institutional, legal, and economic arrangements.

Foreign-Exchange Reserves

External reserve assets held by a monetary authority for purposes including international payments, exchange-rate management, liquidity, and financial stability.

Information Loss

The reduction in information about an underlying system caused by representation, aggregation, measurement, compression, or transformation.

Institution

A durable rule, organization, convention, or structure governing social and economic interaction.

Large Language Model

A large neural-network model trained to represent and generate natural language and related information.

Market Capitalization

The market value of publicly traded equity, conventionally calculated as share price multiplied by shares outstanding.

Money

A socially and institutionally recognized medium of exchange, unit of account, and store of value.

Post-Money Valuation

A valuation of a company immediately following a financing transaction.

Sharpe Ratio

A risk-adjusted performance statistic measuring excess return relative to return volatility.

Substrate

The physical material on which a banknote's printed information and security features are produced.

Technology

A system of knowledge, processes, tools, and organizational practices used to transform resources into useful outputs.

Valuation

An estimate of the economic value of an asset, enterprise, claim, or stream of future cash flows.

Welfare

A broad measure of economic and social well-being, potentially incorporating consumption, income, risk, leisure, stability, and other dimensions.

The End